CUADebug: Diagnosing and Repairing Computer-Use Agent Failures

EMNLP2026 BUDAPEST

Weijia Zhang[1,3§], Kunlun Zhu[1§], Zeyi Liu[1], Yinting Chen[1], Tianyi Ma[1], Jiateng Liu[1], Jiaxun Zhang[1], Bingxuan Li[1], Pan Lu[2], Heng Ji[1†], Jiaxuan You[1†]

[1]University of Illinois Urbana-Champaign [2]Stanford University [3]Yale University [§]Leading authors [†]Co-senior authors {weijia4, kunlunz2, hengji, jiaxuan}@illinois.edu

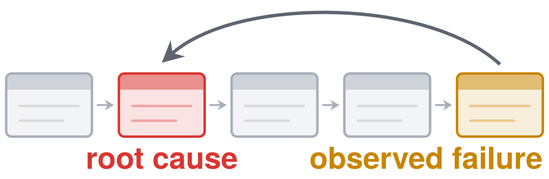
11.1 → 19.4% root cause found

61.77 → 68.14% OSWorld accuracy

Paper | Project | Code | Data

1 Motivation

- Computer-use agents act on screenshots with mouse and keyboard actions.
- When a run fails, the cause is often several steps earlier.



2 CUAErrorBench

204 failed OSWorld runs from 3 agents, each labeled with its root-cause step and error type.

Perception | Grounding | Reasoning | External / System | Others



Reasoning errors dominate; perception and grounding cause 30%.

3 Error Taxonomy

5 families, 30 subtypes

- **Perception** 5 subtypes — *Misread what was visible?*
- **Grounding** 4 subtypes — *Right intent, wrong target?*
- **Reasoning** 13 subtypes — *Wrong plan chosen or kept?*
- **External / System** 7 subtypes — *Environment blocked progress?*
- **Others** 1 subtype — *Infeasible task*

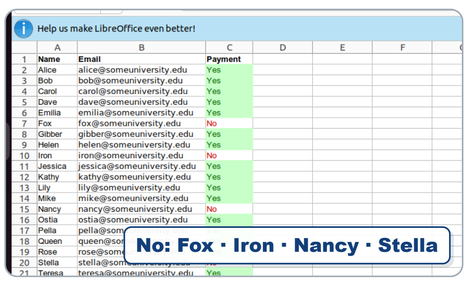
4 A Real Failure

The run fails at **step 12**, but the mistake is made at **step 6**, where the agent lists **Kelly** instead of **Fox**.

REAL CASE **Task (abridged): Find unpaid tuition e-mails and add them to the receiver field**
Environment: LibreOffice Calc → Thunderbird

A Failure trajectory

1 after t=5 Source record



The payment column identifies exactly four unpaid students; row 12 is Kathy with **Yes**.

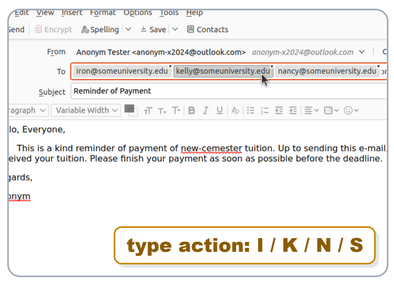
⚠ ROOT CAUSE

2 t=6 Root reasoning



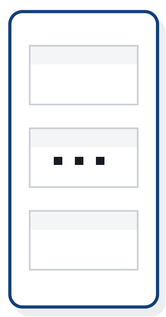
The agent invents **Kelly** at row 12 and drops Fox, even though the visible row is Kathy / Yes.

3 t=10 Recipient entry



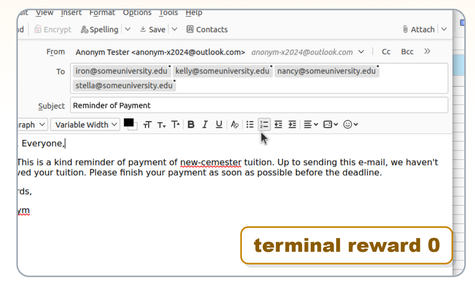
The four-address action carries the false belief into Thunderbird; the complete chips are visible at t=12.

4 1 more action



⚠ OBSERVED FAILURE

5 t=12 Final state



The draft still contains Kelly and still omits Fox when the trajectory declares completion.

5 CUADebugger

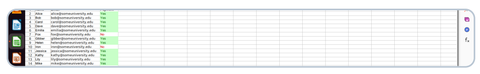
It inspects the before/after screenshots of selected steps, finds the earliest error, and writes a report that guides a new run. For the run above, the re-run **fails without the report (score 0)** and **succeeds with it (score 1)**.

1 Failed trajectory package

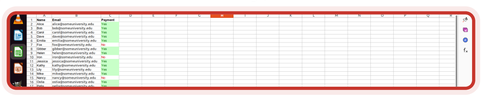
TASK Add unpaid students' e-mails to the receiver field
APPS Calc → Thunderbird

Input: screenshots + reasoning/action trace

- **t=5** payment record



- **t=6 ROOT** invents Kelly; misses Fox



- **t=10** wrong recipients

Iron, Kelly, Nancy, Stella; no Fox

- **t=12 FAIL** declares success; reward 0



INPUT task + trajectory + terminal failure

2 Tool-guided inspection & localization

`get_step_details(6)` + `get_step_details(10)`



before / after | response + action | later evidence

Candidate inspection path

inspect t=6 → compare t=5/6 → inspect t=10 → localize t=6

Root: t=6 reasoning invents Kelly and omits Fox.

OUTPUT root-step candidate + grounded evidence

3 Structured RCA record

Input: candidate step + inspected evidence

ROOT STEP `root_error_step = 6` — Invents Kelly; omits Fox

TAXONOMY `taxonomy_tag = P1` — Visual hallucination

EVIDENCE t=5: Fox = No; t=6: Kelly/no Fox; t=10: wrong list

CORRECTION Use all No rows: Fox, Iron, Nancy, Stella

OUTPUT structured RCA record

4 Debugger-guided re-rollout

Matched continual: Claude 4.5 Sonnet; restart t=51

Continual re-rollout: Recovery / Overall

- Direct continuation: **12.50% / 61.22%**
- Self-Debug: **13.89% / 61.77%**
- Machine RCA: **21.53% / 64.82%**
- Ours + episodic memory: **25.69% / 66.48%**

Takeaway

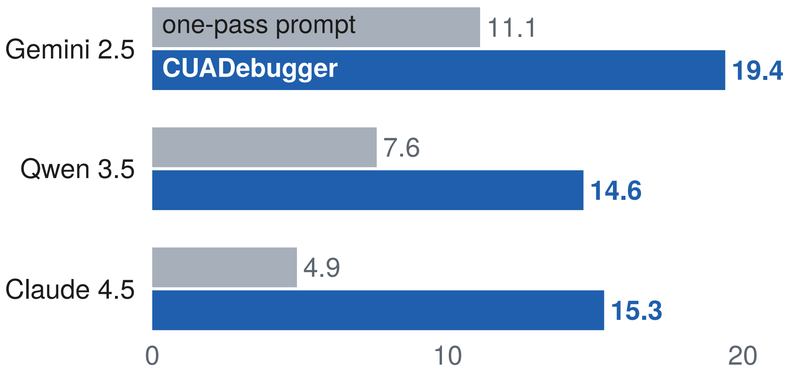
RCA outperforms reflection; episodic memory adds **+4.16 pp recovery (+1.66 pp overall)**.

OUTPUT re-rolled trajectory + outcome

6 Results

Root-cause analysis

type and step both correct (%)



Re-execution

accuracy on 361 OSWorld tasks (%)

	Single	Continual
Baseline	61.77	61.22
Self-debug	62.33	61.77
Machine RCA	67.59	64.82
CUADebug	**68.14**	**66.48**
Human diagnosis	*68.98*	*67.87*

Recovered failures (%)

